Session 12
By: Prof. Raksha Sharma

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Advanced Data Selection and Filtering

Revising and Diving deep to make the data and ourself ready for Data Analysis

Preprocessing: Data Selection and Filtering

- Data selection: Choosing specific rows and columns from a dataset as per our requirement

- Data Filtering: Use operators to focus on certain data of our interest

Both go together

In Real Life

Data	**Our Interest in this part (Filtering)**
HR Department	*Employees with salary > ₹50,000*
Academics	*Students scoring above 9 Points*
Health	*Patients older than 60 years*
E-commerce	*Orders delivered in the last month*
Banking	*More than 5 transaction for a customer in a day*

Data Selcetion mechanisms

- Select a particular column and analyse yourself statestically

  df["Salary"] # select salary column

  df[“Salary”].mean() # compute mean of salary

  df[["Name", "Salary"]]   #Selecting multiple columns

- *Selcting rows using loc*
  Df.loc[row_index]

  Df.loc[1:3]   # Selecting multiple rows

  df.iloc[0:3, 1:3] # Selecting frist three rows with 1st and 2nd column

When *loc* and when *iloc*

loc	iloc
Uses labels	Uses positions
End index included loc[0:2]	End index excluded in iloc[0:2]

Filtering operators

- Greater than/Less than/ Equality/ Not equality

df[df["Salary"] > 60000]

df[df["Department"] == "HR"]

df[df["Department"] != "IT"]

- Logical operators to combine more than two filters in one instruction

df[(df["Department"] == "IT") & (df["Salary"] > 40000)]

More refined arrangement of the data

- Sorted data on a particular column
    - Sort students marks in ascending/descending order to see the break points in marks manually

      df.sort_values(by="Marks", assessing = False) # default is ascending

    - Sorting on multiple columns - Get sorted marks for each sub-batch of students

      df.sort_values (by=[“Marks”, “sub-batch”])

      df.sort_values(by=["Department", "Salary"], ascending=[True, False]) #ascending on first column and descending on second column

Summary of functions for selection and filtering

Select single column	df["column"]
Select multiple columns	df[["col1","col2"]]
Select rows by label	df.loc[]
Select rows by index	df.iloc[]
Greater than filter	df[df["col"] > value]
Equality filter	df[df["col"] == value]
AND condition	&
OR condition	|
Ascending sort	sort_values()
Descending sort	sort_values(ascending=False)
Multi-column sort	sort_values(by=[...])
Reorder columns	df[[new_order]] # new order of old attributes, just by giving names of attributes

Thank you!